

5.2 - An Introduction to Graphs

Problem Solving, Math 1000

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Question

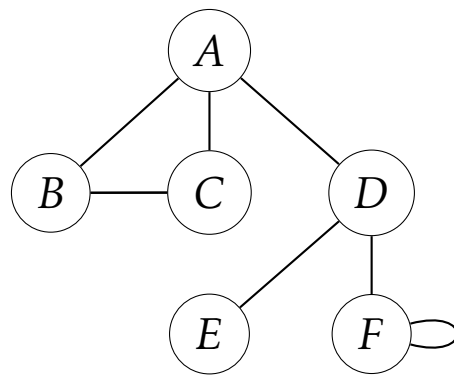
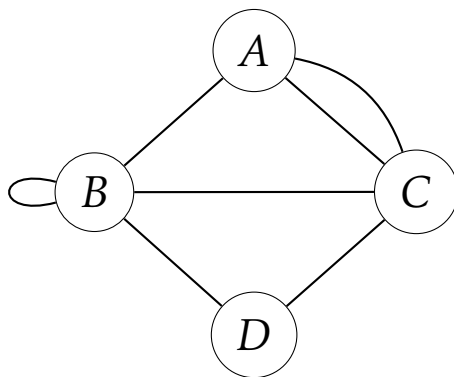
How can we represent street routing problems, or problems such as the bridges of Konigsberg as efficiently as possible? That is to say, how can we talk about "only the important parts" of the setup to these problems?

Definition

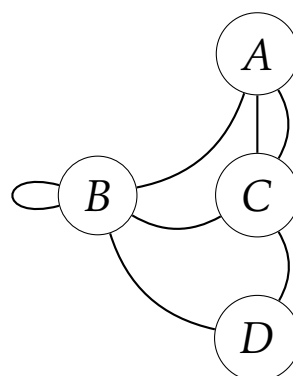
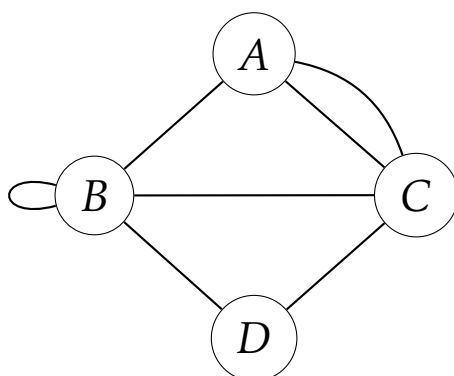
A **graph** is a set of vertices and a set of edges. In very simple terminology, a graph is just a generalization of a set of "things" and a set of "relationships" between them. We draw these as dots (vertices) with lines (edges) between some, all, or none of the dots.

Example

The following are some examples of graphs:



Note: we are not concerned with the placement of the vertices or the way in which the edges are drawn – the only important thing is which vertices there are and which edges exist between them. For example, the following graphs are identical from our perspective:

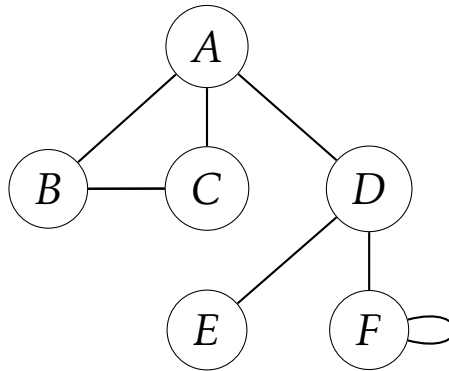


Definition

Given a graph G , we define the **vertex set** of the graph G , denoted $V(G)$, to be the set of vertices in the graph.

Example

The vertex set of the graph G given below is: $V(G) = \{A, B, C, D, E, F\}$

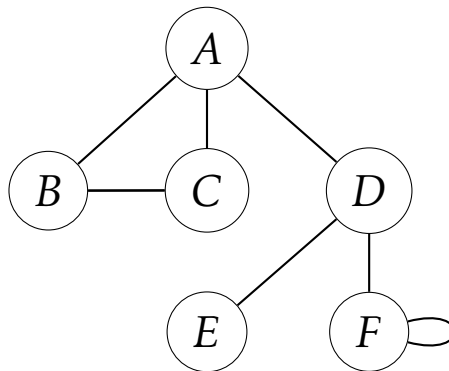


Definition

Given a graph G , we define the **edge set** (or, as some call it, the **edge list**) of the graph G , denoted $E(G)$, to be the set of edges in the graph. We denote an edge between the vertex A and the vertex B by AB (or, equivalently, by BA).

Example

The vertex set of the graph G given below is: $E(G) = \{AB, AC, BC, AD, DE, DF, FF\}$

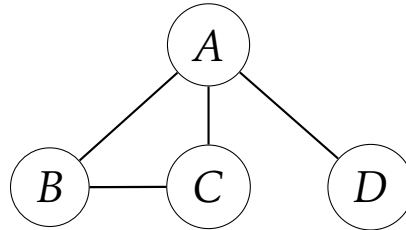


Definition

If two vertices in a graph are connected by an edge, we say that they are **adjacent**. More formally, that is to say that, for a given graph G , if vertex A and vertex B are elements of $V(G)$ and if AB is an element of $E(G)$, then vertex A is adjacent to vertex B and vertex B is adjacent to vertex A .

Example

In the graph given below, vertex A is adjacent to vertices C , B , and D , vertex B is adjacent to vertices A and C , vertex C is adjacent to vertices B and A , and vertex D is adjacent to vertex A . Note that neither vertex B nor C is adjacent to vertex D .

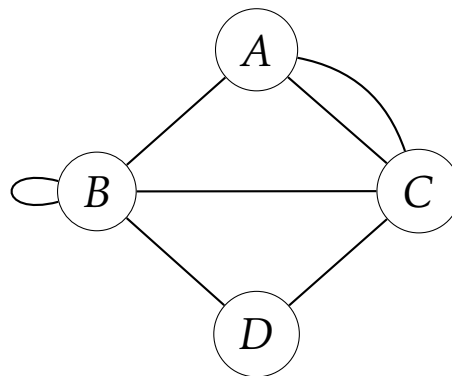


Definition

An edge from a vertex to itself is called a **loop**.

Example

In the graph below, vertex B has a loop.

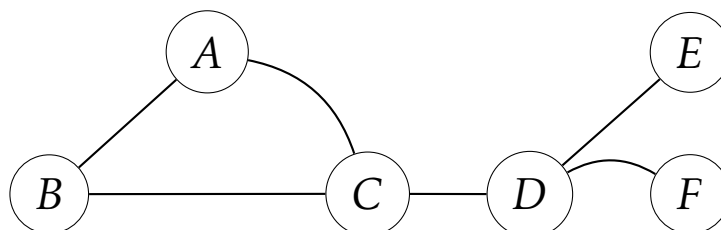


Definition

The number of edges that meet at a vertex is called the **degree** of the vertex. For some vertex (let's call it A for now) with n edges meeting at it, we write $\deg(A) = n$. If the degree of a vertex is an odd number, we call the vertex an **odd vertex**. If the degree of a vertex is an even number, we call the vertex an **even vertex**.

Example

In the graph below, $\deg(A) = 2$, $\deg(B) = 2$, $\deg(C) = 3$, $\deg(D) = 3$, $\deg(E) = 1$, and $\deg(F) = 1$.

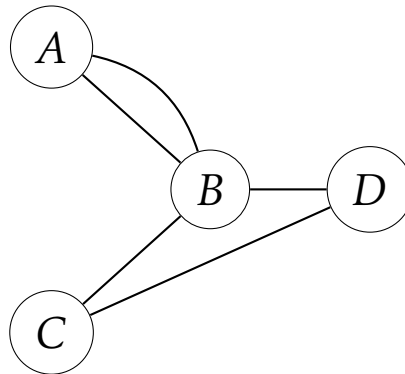


Definition

If there is more than one edge connecting two vertices in a graph, we say that the graph has **multiple edges**.

Example

The following graph has more than one edge connecting vertex A to vertex B , and therefore has multiple edges.

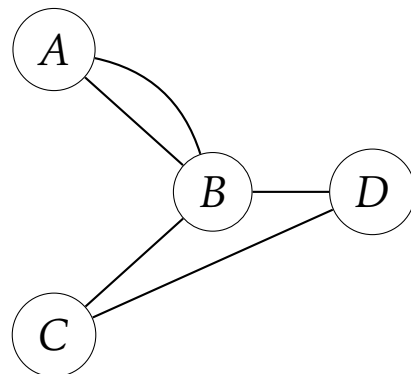
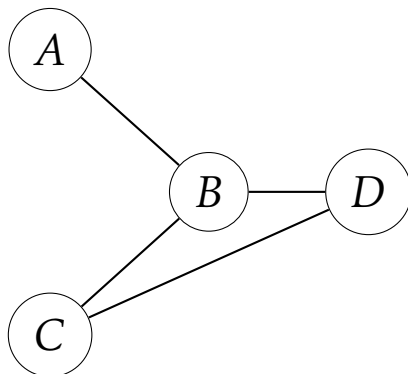


Definition

A **simple graph** is a graph without loops or multiple edges.

Example

The first graph below is simple, while the second graph is not.

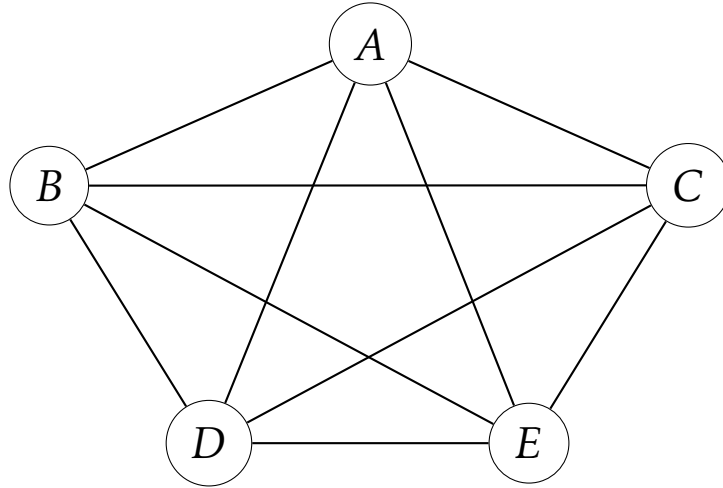


Definition

A **clique**, or **complete graph**, is a simple graph with as many edges as possible.

Example

The following graph is complete.

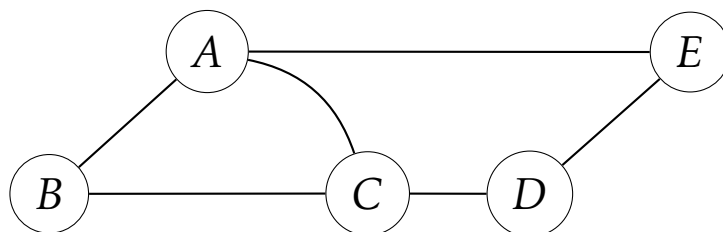


Definition

A sequence of edges each adjacent to the next, with no edge included more than once, that starts and ends at different vertices is called a **path**. A sequence of edges each adjacent to the next, with no edge included more than once, that starts and ends at the same vertex is called a **circuit**. The number of edges in such a sequence is called the **length** of the path or circuit.

Example

The sequence of adjacent edges AB, BC, CD in the graph below is a path with length 3, while AB, BC, CD, DE, EA is a circuit with length 5. There are more paths and circuits in the graph below, these are just two examples.

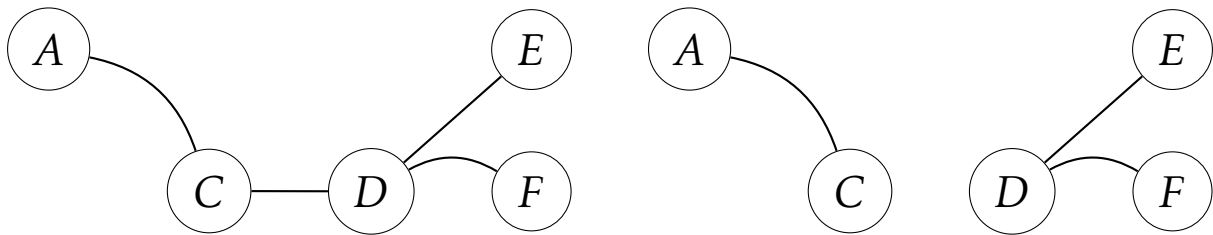


Definition

A graph which is not "made of a single piece" is called **disconnected**. More formally, a graph is **disconnected** if there is not at least one path between any two pairs of vertices in the graph. If a graph is not disconnected, we say it is **connected**.

Example

The first graph below is connected, while the second is not.

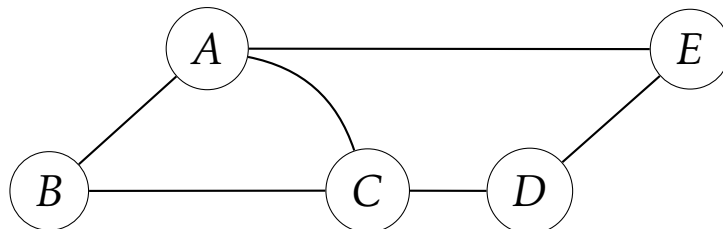


Definition

A **Euler path** is a path which covers all the edges of a graph. A **Euler circuit** is a circuit which covers all the edges of a graph.

Example

The path AB, BC, CA, AE, ED, DC in the graph below is a Euler path.



Definition

In a *connected* graph, a **bridge** is an edge that would disconnect the graph if deleted.

Example

In the first graph below, the edge CD is a bridge. In the second graph, there are no bridges because this graph is not connected to begin with.

